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Entropy-Guided K-Means Clustering

Project Report

Course – MCSE (AI & DS)

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1 Entropy-Guided K-Means Clustering

1.1 Problem Description

Clustering is a fundamental unsupervised learning task aimed at grouping similar data points without prior labels. One of the most widely used clustering techniques is K-means, which requires the number of clusters k to be specified in advance. However, selecting an optimal value of k remains a challenging problem and significantly affects clustering performance.

In this work, we implement an entropy-guided approach to determine the optimal number of clusters for the K-means algorithm. The method is based on computing probabilistic cluster memberships using a distance-based softmax function and evaluating the uncertainty (entropy) associated with cluster assignments. For each candidate value of k, the normalized entropy of the dataset is calculated, and the optimal number of clusters is selected as the value that minimizes this normalized entropy.

The proposed method is evaluated on the *Credit Card Dataset for Clustering* obtained from Kaggle. To assess its effectiveness, the results are compared with two commonly used approaches:

- Standard K-means using the Elbow Method
- Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

The performance of all methods is evaluated using the following clustering quality metrics:

- Silhouette Score
- Davies-Bouldin Index

The objective of this study is to analyze whether entropy-based selection of k provides a more reliable alternative compared to traditional methods.

1.2 Dataset Setup

1.2.1 Dataset Description

The dataset used in this study is the *Credit Card Dataset for Clustering*, which is widely applied for customer segmentation tasks in financial analytics.

The dataset contains information about approximately 9000 active credit card holders over a period of six months. Each record represents a single customer and includes 18 behavioral features describing transaction frequency, purchase patterns, payment behavior, credit utilization, and tenure. These attributes provide valuable insights into customer usage habits and support meaningful clustering analysis.

The dataset is structured at the customer level and includes both monetary and frequency-based variables that reflect different aspects of credit card usage. The detailed description of the variables used in this study is given below:

- **CUST_ID**: Unique identification number of the credit card holder (categorical feature).
- **BALANCE**: Remaining balance in the account available for purchases.
- **BALANCE_FREQUENCY**: Frequency of balance updates (range: 0 to 1).
- **PURCHASES**: Total amount of purchases made from the account.
- **ONEOFF_PURCHASES**: Maximum purchase amount made in a single transaction.
- **INSTALLMENTS_PURCHASES**: Amount spent through installment-based purchases.
- **CASH_ADVANCE**: Cash advance amount taken by the customer.
- **PURCHASES_FREQUENCY**: Frequency of purchases (range: 0 to 1).
- **ONEOFFPURCHASESFREQUENCY**: Frequency of one-time purchases.
- **PURCHASESINSTALLMENTSFREQUENCY**: Frequency of installment purchases.
- **CASHADVANCEFREQUENCY**: Frequency of cash advance usage.
- **CASHADVANCETRX**: Number of cash advance transactions.
- **PURCHASES_TRX**: Number of purchase transactions.
- **CREDIT_LIMIT**: Credit limit assigned to the customer.
- **PAYMENTS**: Total payment amount made by the customer.
- **MINIMUM_PAYMENTS**: Minimum payment amount made by the customer.
- **PRCFULLPAYMENT**: Percentage of full payments made by the customer.
- **TENURE**: Duration (in months) of credit card service usage.

The analysis of this dataset enables the identification of distinct groups of credit card users based on their spending behavior and financial activity patterns.

1.2.2 Preprocessing

Before applying clustering algorithms, several preprocessing steps were performed to improve data quality and ensure reliable clustering performance. Since K-Means is a distance-based algorithm, proper feature scaling and outlier handling are essential for obtaining meaningful clusters.

Handling Missing Values: The dataset contained a small number of missing values in some attributes such as *MINIMUM_PAYMENTS* and *CREDIT_LIMIT*. These missing entries were removed using row-wise deletion (`dropna`) to maintain dataset consistency and avoid distortion in distance calculations.

Log Transformation: Several financial variables such as *BALANCE*, *PURCHASES*, *CASH_ADVANCE*, and *CREDIT_LIMIT* exhibited strong right-skewed distributions. To reduce skewness and stabilize variance, a logarithmic transformation (`log1p`) was applied. This helped improve cluster separability and reduced the dominance of large-value observations.

Outlier Treatment (Winsorization): Extreme values can significantly affect centroid-based clustering algorithms. Therefore, Winsorization was applied by clipping feature values at the 1st and 99th percentiles. This preserved the overall distribution while limiting the influence of outliers.

Feature Normalization: Since the dataset contains variables measured on different scales (e.g., transaction counts, monetary values, and frequency scores), Z-score standardization was applied to all numerical features. Each feature was transformed to have zero mean and unit variance using:

$$z = \frac{x - \mu}{\sigma}$$

This ensured that all variables contributed equally to the distance computation during clustering.

These preprocessing steps improved numerical stability, reduced noise and skewness, and enhanced the effectiveness of the entropy-guided K-Means clustering algorithm.

1.3 Algorithms & Formulas

1.3.1 K-Means Clustering

The objective of the K-Means algorithm is to partition a dataset into K clusters such that the total intra-cluster variance is minimized.

The algorithm works iteratively through the following steps:

1. Select the number of clusters K.
2. Initialize K centroids randomly or using the K-Means++ initialization method.
3. Assign each data point to the nearest centroid based on Euclidean distance.
4. Update each centroid by computing the mean of all data points assigned to that cluster.
5. Repeat the assignment and update steps until convergence is achieved (i.e., centroids no longer change significantly).

The objective function minimized by K-Means is:

$$J = \sum_{i=1}^{n} \min_{1 \leq j \leq K} \|x_i - \mu_j\|^2 \tag{1.1}$$

where x_i represents the ith data point and μ_j represents the centroid of the jth cluster.

K-Means is efficient and simple to implement, but its performance depends heavily on selecting an appropriate value of K, which motivates the use of entropy-based cluster evaluation methods.

1.3.2 Entropy Concept

Entropy is an information-theoretic measure that quantifies the uncertainty or randomness in a probability distribution. In clustering, entropy can be used to measure how confidently data points belong to particular clusters.

The entropy of a random variable X is defined as:

$$H(X) = -\sum_{i=1}^{n} p(x_i) \log p(x_i) \tag{1.2}$$

where $p(x_i)$ represents the probability of occurrence of event x_i.

In the context of clustering, lower entropy indicates clearer separation between clusters, whereas higher entropy suggests ambiguity in cluster assignments. Therefore, entropy can serve as a useful metric for evaluating clustering quality.

1.3.3 Entropy-Guided Modification

In the proposed entropy-guided K-Means approach, entropy is incorporated as a criterion to determine the optimal number of clusters by measuring the uncertainty in cluster assignments.

Instead of assigning each data point strictly to one cluster (hard assignment), a soft assignment probability is computed using a probability distribution controlled by a parameter β:

$$P_{ij} = \frac{\exp(-\beta \|x_i - \theta_j\|^2)}{\sum_{j'} \exp(-\beta \|x_i - \theta_{j'}\|^2)} \tag{1.3}$$

where P_{ij} represents the probability that data point x_i belongs to cluster j, and θ_j represents the centroid of cluster j.

Using these probabilities, the entropy for each data point is calculated as:

$$H_i = -\sum_{j=1}^{K} P_{ij} \log P_{ij} \tag{1.4}$$

The average entropy across all data points is then computed and normalized as:

$$H_{norm} = \frac{1}{n} \sum_{i=1}^{n} \frac{H_i}{\log K} \tag{1.5}$$

The optimal number of clusters is selected as the value of K that minimizes the normalized entropy. A lower entropy value indicates more confident cluster assignments and better cluster separation.

To further improve clustering performance, the parameter β is optimized using Simulated Annealing, which helps identify the best soft assignment structure and enhances the robustness of cluster selection.

1.4 Hyperparameter Tuning

Proper selection of hyperparameters plays an important role in improving clustering performance and ensuring meaningful segmentation results. In this study, several important parameters of the entropy-guided K-Means algorithm were carefully tuned as described below.

- **Number of Clusters (k):**

  Selecting the optimal number of clusters is one of the main challenges in K-Means clustering. Instead of choosing k manually or relying only on heuristic methods such as the Elbow Method, this work determines the optimal value of k using normalized entropy. The algorithm evaluates clustering performance for values of k in the range of 2 to 29 and selects the value that minimizes the normalized entropy:

  $$k^* = \arg\min_k \left(\frac{H_k}{\log k} \right)$$

  where H_k represents the average entropy of cluster assignments for a given k. A lower entropy value indicates clearer separation between clusters.

- **Initialization Method:**

  Centroid initialization significantly affects convergence speed and clustering quality. In this study, the K-Means++ initialization technique was used instead of random initialization. K-Means++ selects initial centroids that are well separated from each other, which improves stability, reduces the chance of poor local minima, and accelerates convergence.

- **Maximum Iterations:**

  The maximum number of iterations controls how long the algorithm updates cluster centroids before stopping. A limit of 50 iterations was used in this implementation, which was sufficient for convergence due to the use of K-Means++ initialization and standardized feature scaling. Additionally, convergence was checked using centroid movement with a stopping threshold of 10^{-3}.

- **Entropy Threshold and Beta Optimization:**

  Instead of using a fixed entropy threshold, clustering quality was evaluated using normalized entropy values computed from soft assignment probabilities. The parameter β, which controls assignment sharpness, was automatically optimized using Simulated Annealing. The optimization process started with an initial value $\beta = 0.1$ and gradually refined it over multiple iterations to minimize entropy.

This adaptive tuning of β improves clustering robustness by ensuring that cluster assignments remain neither overly rigid nor excessively uncertain. The final optimal number of clusters was selected based on the minimum median normalized entropy obtained across the best-performing β values.

Overall, these hyperparameter tuning strategies ensured stable convergence, improved cluster separability, and enhanced the reliability of the entropy-guided clustering framework.

1.5 Comparison with Other Models

To evaluate the effectiveness of the proposed entropy-guided K-means approach, a comparative analysis is performed with two widely used clustering methods:

- K-means with the number of clusters determined using the Elbow Method
- Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

The Elbow Method determines the optimal number of clusters k by analyzing the variation of within-cluster sum of squares (inertia) and identifying the point where the rate of decrease sharply changes. While simple and intuitive, this method often suffers from ambiguity in identifying the exact elbow point.

DBSCAN, on the other hand, is a density-based clustering algorithm that groups data points based on density connectivity and is capable of identifying noise or outliers.

The proposed entropy-guided K-means method introduces a probabilistic framework for cluster assignment and selects the optimal number of clusters by minimizing the normalized entropy. This provides a more data-driven and theoretically grounded alternative to heuristic approaches like the Elbow Method.

1.5.1 Evaluation Metrics

The performance of all clustering methods is evaluated using the following metrics

- **Silhouette Score:**

  The Silhouette Score measures how similar a data point is to its own cluster compared to other clusters. It ranges between -1 and 1, where higher values indicate better clustering performance. A high silhouette score suggests that clusters are well separated and internally cohesive.

  Mathematically, the silhouette coefficient for a data point is defined as:

  $$S = \frac{b - a}{\max(a, b)}$$

  where a represents the average intra-cluster distance and b represents the average distance to the nearest neighbouring cluster.

- **Davies-Bouldin Index:**

  The Davies-Bouldin Index evaluates clustering quality based on the ratio of within-cluster scatter to between-cluster separation. Lower values indicate better clustering performance, meaning clusters are compact and well separated.

  It is defined as:

$$DB = \frac{1}{K} \sum_{i=1}^{K} \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d(\mu_i, \mu_j)} \right)$$

  where σ_i represents the average distance of points within cluster i to centroid μ_i, and $d(\mu_i, \mu_j)$ represents the distance between centroids of clusters i and j.

These evaluation metrics collectively provide a reliable framework for comparing clustering algorithms and validating the effectiveness of the entropy-guided K-Means approach.

1.6 Results and Discussion

1.6.1 Preprocessing Analysis

The dataset was preprocessed using missing value handling, log transformation, winsorization, and standardization. The boxplot in Figure 1.1 shows that the features are well-scaled with reduced impact of outliers, making the data suitable for clustering.

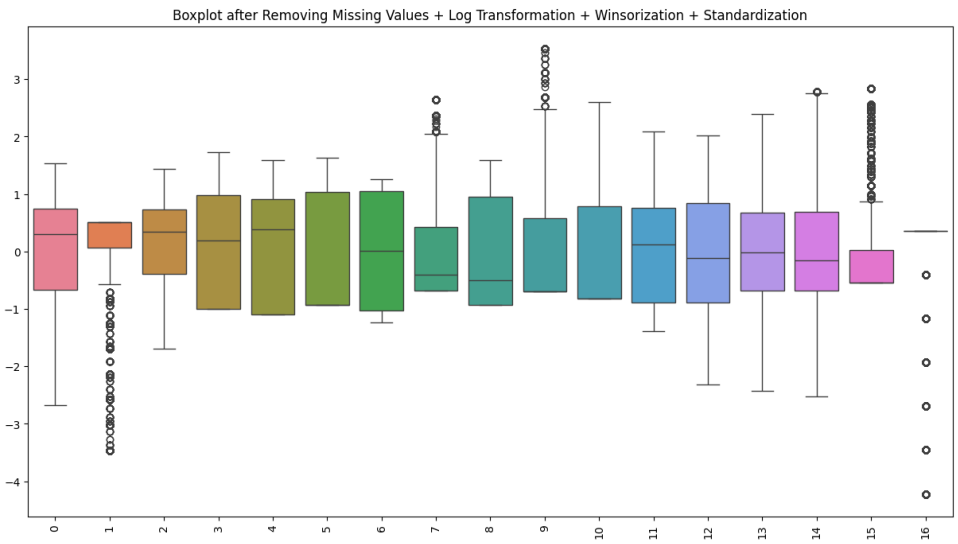

Figure 1.1: Feature distribution after preprocessing

1.6.2 Entropy-Based Cluster Selection

The normalized entropy plots (Figures 1.2 and 1.3) illustrate how entropy varies with different values of k. The entropy-guided method identifies $k = 7$ as the optimal number of clusters by minimizing normalized entropy.

Additionally, the use of simulated annealing for optimizing the parameter β improves the stability of entropy estimation, as shown in Figure 1.4.

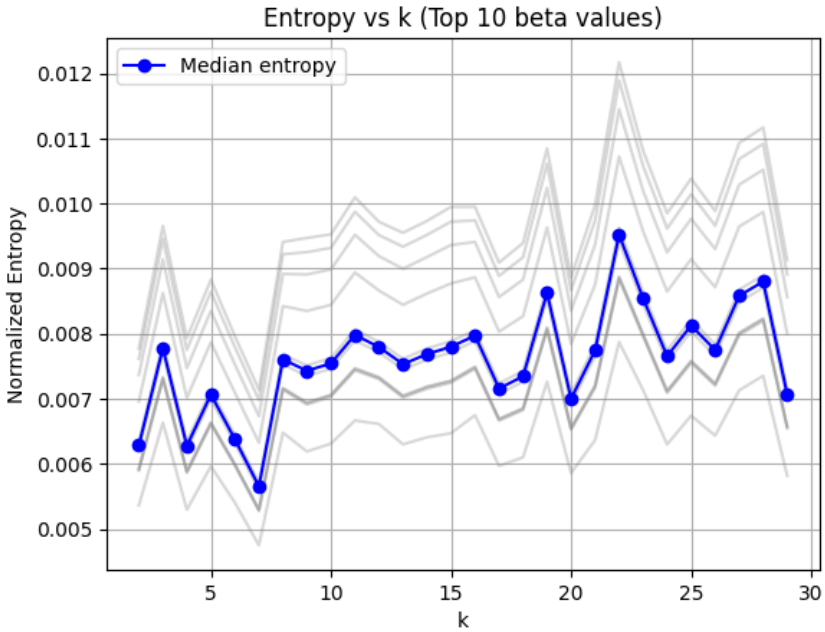

Figure 1.2: Normalized Entropy vs Number of Clusters (k)

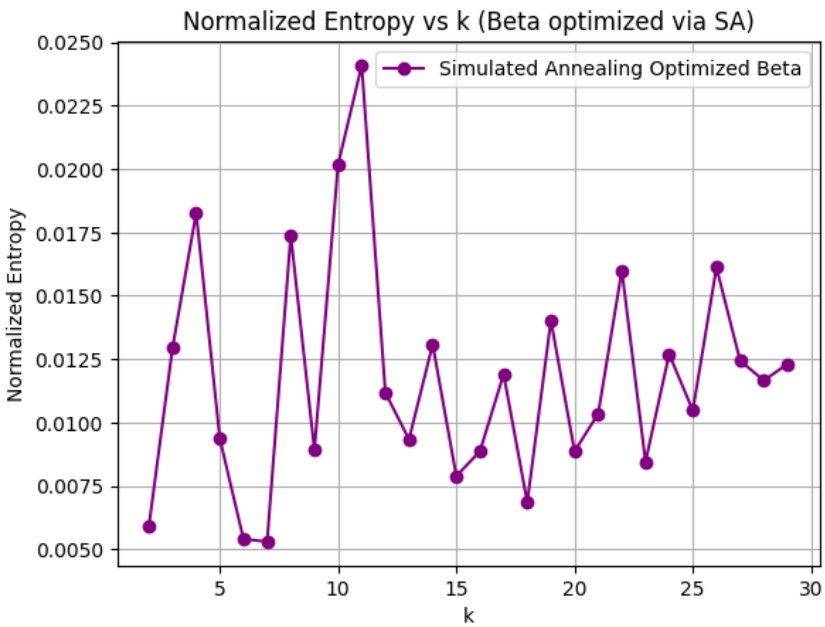

Figure 1.3: Normalized Entropy vs k for optimal β values

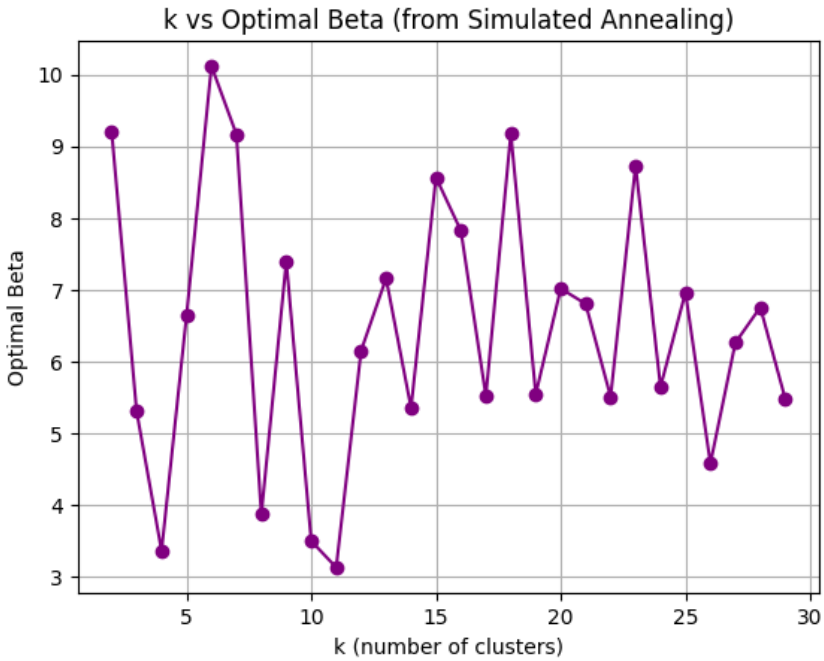

Figure 1.4: Optimal β values for different k

1.6.3 Elbow Method Analysis

The Elbow Method plot (Figure 1.5) shows a gradual decrease in inertia with increasing k, with no sharply defined elbow point. This indicates ambiguity in selecting the optimal number of clusters using this method. However, $k = 9$ was selected as a reasonable estimate.

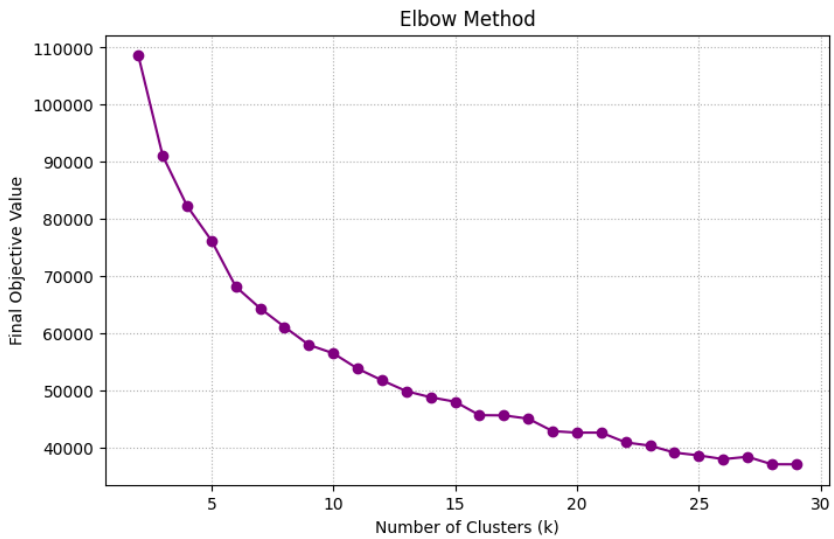

Figure 1.5: Elbow method for determining optimal k

1.6.4 Clustering Visualization

The clustering results for $k = 7$ are visualized using PCA and UMAP in Figures 1.6 and 1.7. The clusters appear well-separated with meaningful groupings, indicating that the

entropy-based selection leads to coherent cluster structures.

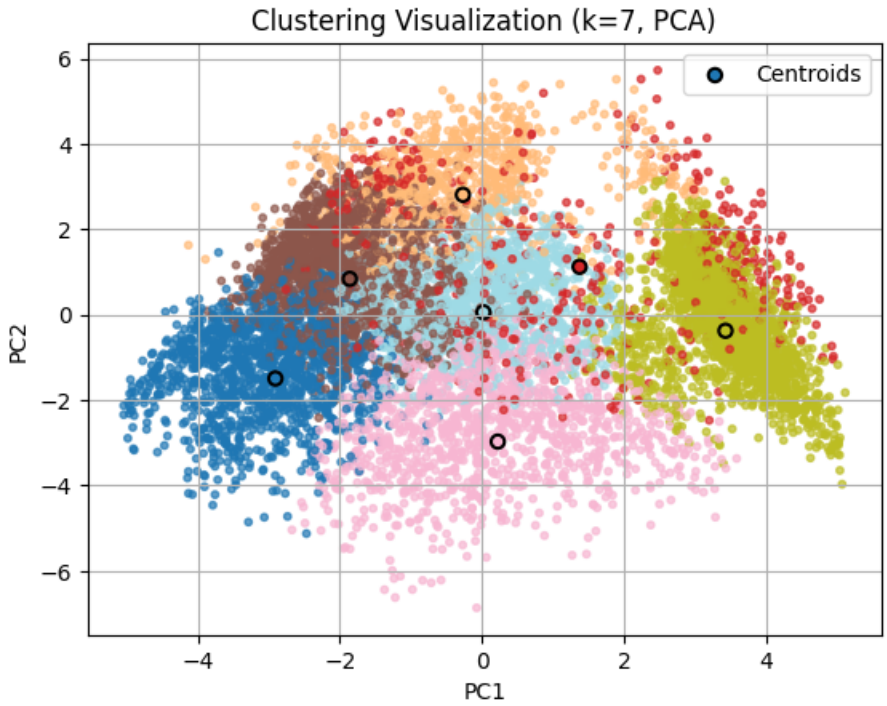

Figure 1.6: Clustering visualization using PCA (k = 7)

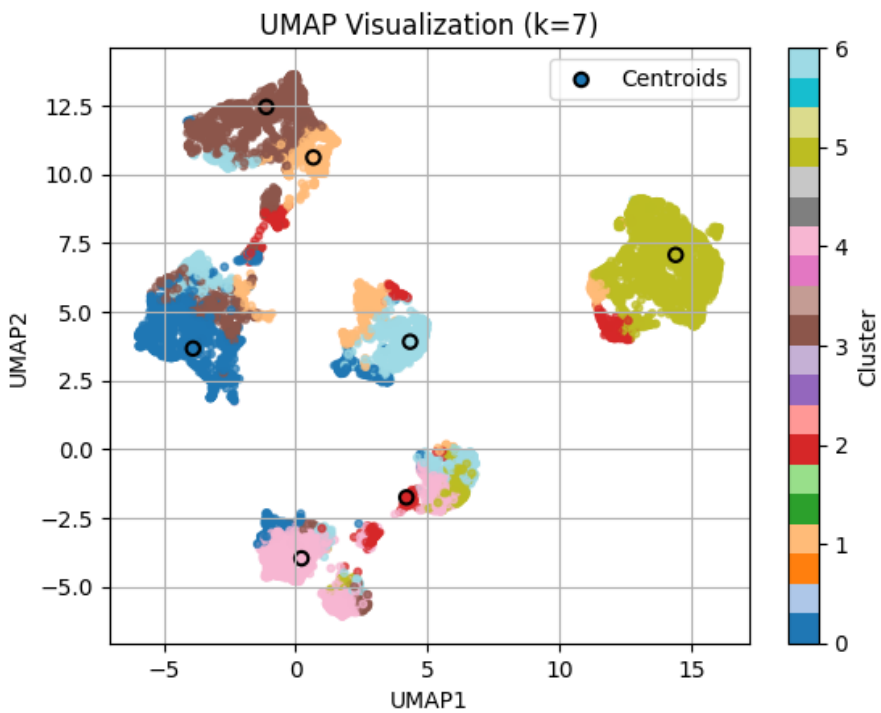

Figure 1.7: Clustering visualization using UMAP (k = 7)

1.6.5 Model Comparison

Table 1.8 summarizes the performance of the different clustering methods.

	Method	Silhouette Score ↑	Davies-Bouldin Index ↓
0	K-means (Elbow k=9)	0.1853	1.4986
1	K-means (Entropy k=7)	0.2209	1.3929
2	DBSCAN	-0.1105	1.0133

Figure 1.8: Comparison of clustering methods

From the results:

- The entropy-guided K-means achieves the highest Silhouette Score (0.2209), indicating better cluster cohesion and separation compared to the Elbow-based K-means (0.1853).
- DBSCAN achieves the lowest Davies-Bouldin Index (1.0133), suggesting better cluster separation in terms of this metric. However, it performs poorly in terms of Silhouette Score (-0.1105), indicating unstable or poorly defined clusters.
- The entropy-based method provides a balanced improvement, achieving both higher silhouette score and lower Davies-Bouldin index than standard K-means.

1.6.6 Summary

The results demonstrate that the entropy-guided K-means approach provides a more reliable method for selecting the optimal number of clusters compared to the Elbow Method. By minimizing uncertainty in cluster assignments, the method produces more coherent and well-separated clusters.

Although DBSCAN performs well in terms of the Davies-Bouldin Index, its negative Silhouette Score indicates poor cluster consistency for this dataset, likely due to varying density and high dimensionality.

Overall, the entropy-based approach offers a data-driven and robust alternative for cluster selection, leading to improved clustering performance in comparison to traditional methods.